

Stock-Market Dynamics, Revision 6

The Winners: the Systematic Buffett Screen
and the Validated Regime Engine

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1 Introduction

Seven pre-registered return-forecast mechanisms were tested against 2000–2022 S&P 500 data. **All seven failed walk-forward validation.** This revision removes their sections and figures from the narrative and rebuilds the report around the two things that survived:

1. **Winner — the systematic Buffett screen (§3).** A quality/value/low-beta stock screen implementing the unlevered selection leg of “Buffett’s Alpha” [1], paper-backtested 2009–2022: **CAGR 14.8% vs. 13.6% SPY, Sharpe 1.27 vs. 0.94**, FF3 alpha +0.46%/month ($t = 2.71$). It is the only return-generating mechanism in this project with a positive validated track record — but it was **not** pre-registered, so it enters the paper-trading protocol as a promising backtest, not a proven edge. No real capital until the paper record holds.
2. **Validated, not a return forecaster — the regime capital-distribution engine (§4).** The 24-state capital census, conditional scenario engine, and concentration/fragility monitor on clean data. Its verdict is unchanged since rev-4 and re-confirmed in rev-5: it prices *states*, not *returns*. A diagnostic and stress-testing instrument, never a deployment signal.

The failed candidates are not all refuted theories — several are weak-evidence non-replications with real design limitations — but none of them forecasts returns in our data, so they leave the narrative. §5 gives each its one-line reason; the full test documents remain in the repository as honest negatives.

2 Related work

Buffett’s Alpha. Frazzini, Kabiller & Pedersen [1] decompose Berkshire Hathaway’s returns into exposures to value (HML), quality (QMJ: profitability, growth, safety, payout) and betting-against-beta (BAB), levered $\sim 1.6\times$ through Berkshire’s insurance float. §3 implements the unlevered stock-selection leg. Factor series come from the Ken French Data Library [2].

Tested and discarded. Brown, Howard & Lundblad [3] find high-minus-low hedge-fund crowdedness earns +2.8%/yr value-weighted ($t = 6.2$) — a compensated risk premium, not a mispricing to fade. Our public-13F proxy does not replicate it in the large-cap panel (§5). Rapach, Ringgenberg & Zhou [4] report a negative relationship between aggregate short interest and market returns ($R^2 \approx 13\%$ annual in-sample); the sign reverses in our modern subsample. Greenwood, Shleifer & You [5] find industry run-up characteristics (acceleration, rising volatility, issuance, new-firm tilt) forecast crash risk; our individual-stock adaptation does not. Lou & Polk [6] find comomentum predicts lower momentum returns; our momentum-lane comomentum has the wrong sign out of sample. The Goyal–Santa-Clara dispersion result does not replicate in the large-cap panel. Daniel & Moskowitz [7] characterize momentum crashes as a loser-optionality “panic state” — a mechanism we never turned into a working forecast.

3 Winner: the systematic Buffett screen

3.1 Design

Universe: true historical S&P 500 members each June (`pane1_v2.csv`, clean post-unit-fix data of §4.2), rebalances June 2009–2021. Prices and membership are **measured** [9, 8]: no survivorship bias by construction.

Table 1: Buffett screen: score components.

Component	Formula	Score
Quality	ROE = NetIncomeLoss/StockholdersEquity (FY); ROA secondary	mean($z(\text{ROE})$, $z(\text{ROA})$)
Value	B/M = Equity/mcap(June); E/P = NetIncome/mcap(June)	mean($z(\text{B/M})$, $z(\text{E/P})$)
Low beta	trailing 252-day market beta vs. SPY (≥ 126 obs)	$-z(\text{beta})$
Composite = mean of the three component z -scores.		

Scores (cross-sectional z -scores, winsorized at ± 3 , point-in-time only):

Portfolio: top 30 by composite, equal-weighted, rebalanced **annually at end of June** — Buffett holds for years; monthly churn would be unfaithful to the method. Negative book equity excluded. Delisted holdings exit at the last available price with proceeds redistributed (**assumed**). The June- Y rebalance uses the latest fiscal year ending ≥ 120 days before June 30 Y (10-K filed by then in practice) — the Fama–French end-date convention; SEC companyfacts `filed` dates are unreliable pre-2010 (restated facts carry the restating filing’s date, verified on AAPL FY2007), so the end-date rule governs point-in-time status [10].

Data provenance. Fundamentals (NetIncomeLoss, StockholdersEquity, Assets): **measured** (SEC EDGAR XBRL companyfacts, 10-K facts only, USD [10]); 584/585 tickers fetched, 19 absurd values rejected by absolute-anchor bounds. Coverage: XBRL mandate began 2009 (FY2008+: 424–569 tickers); the June-2008 rebalance had only 3 eligible stocks and was **skipped** — the backtest starts June 2009. 11 tickers (e.g. XOM, PNC, BSX) have 10-Q-only facts and are excluded: a documented gap, not filled. FF factors: **measured** (Ken French Data Library, 202608 CRSP vintage [2]); AQR’s QMJ/BAB downloads are bot-walled (verified 2026-09-28) — not used, no proxy substituted.

3.2 Results (2009-07 to 2022-06, 156 months)

Table 2: Buffett screen vs. SPY, 2009-07–2022-06.

	Buffett screen	SPY
CAGR	14.8%	13.6%
Sharpe (excess, ann.)	1.27	0.94
Ann. volatility	11.1%	14.2%
Max drawdown	−19.6%	−20.0%
Growth of \$1	\$6.01	\$5.26
Calendar-year hit rate	7/14	—
One-way annual turnover	53%	—

FF3 regression (excess returns, HC1 SE, $R^2 = 0.66$): alpha +0.46%/mo ($t = 2.71$), +5.5% annualized; Mkt-Rf **0.64** ($t = 15.3$); SMB −0.31 ($t = -4.5$, large-cap); HML +0.00 ($t = 0.06$).

It wins the down years (2011 **+16.8** / **+1.9**, 2018 **+1.9** / **−4.6**, 2022 **−7.5** / **−20.0**) and lags raging bulls (2020 **−0.6** / **+18.3**, missing mega-cap tech — as Buffett himself did). Holdings pass the smell test (MO, CL, PEP, BRK-B, WMT, MCD, KMB across years). The notable analytical finding: the HML loading is **zero** — the quality component (high ROE) neutralizes the value tilt, so this behaves as a **quality + low-beta** screen, not deep value.

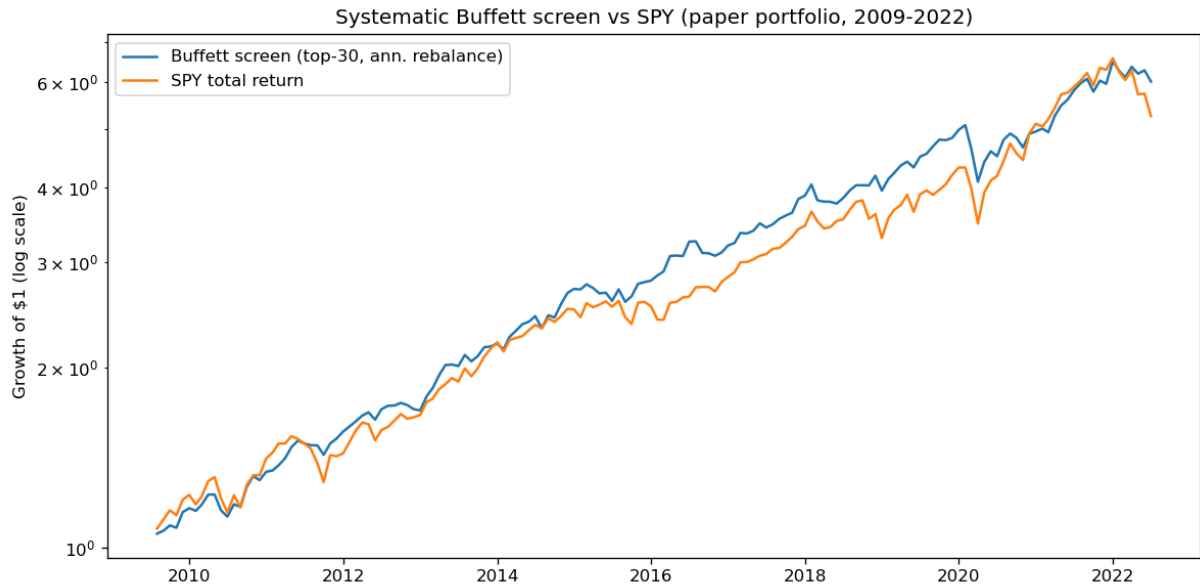


Figure 1: Growth of \$1: systematic Buffett screen (blue) vs. SPY (gray), 2009-07–2022-06. Annual end-of-June rebalance, top-30 equal weight, unlevered.

3.3 Limitations and paper-trading status

Not pre-registered — design choices (top 30, equal weight, June rebalance, composite formula) are standard-academic but were the builder's, so researcher degrees of freedom apply: weaker evidence than the pre-registered negative tests. No float leverage; no crisis deal-flow; no QMJ/BAB attribution; 11 tickers excluded; no transaction costs modeled (53% annual one-way turnover $\approx 0.1\text{--}0.3\%/yr$ at institutional rates, **assumed**); one 14-year window dominated by the post-2009 bull market; 2020 shows the known failure mode.

The screen enters the **paper-trading protocol**: frozen design, live tracking vs. SPY. **No real capital until the paper record holds**; then a small pilot allocation.

4 Validated: the regime capital-distribution engine

4.1 Master equation and reaction graph

Capital is distributed over 24 states (3 momentum $\times$ 2 volatility $\times$ 2 turnover $\times$ 2 size), capital-weighted, on true historical membership. The state vector obeys the cluster master equation

$$\frac{dc}{dt} = \mathbf{P} + \mathbf{S}\mathbf{J} - \mathbf{D}c, \quad (1)$$

with c the 24-bin capital shares, $\mathbf{P}$ the index-entry source, $\mathbf{S}\mathbf{J}$ the reaction-flux stoichiometry, and $\mathbf{D}c$ the exit sink. Figure 2 shows the full reaction-admissibility graph: momentum advection (red/blue, mechanical, $\kappa = 1.0$) and volatility drift (gray dashed, measured 24×24 T).

4.2 Data: the share-unit integrity fix

Filed share counts carried consistent $10^3\text{--}10^6 \times$ unit errors the jump-based cleaner missed — including a backfill reading Sempra Energy at **\$5,000 trillion**, flipping the measured top-lane

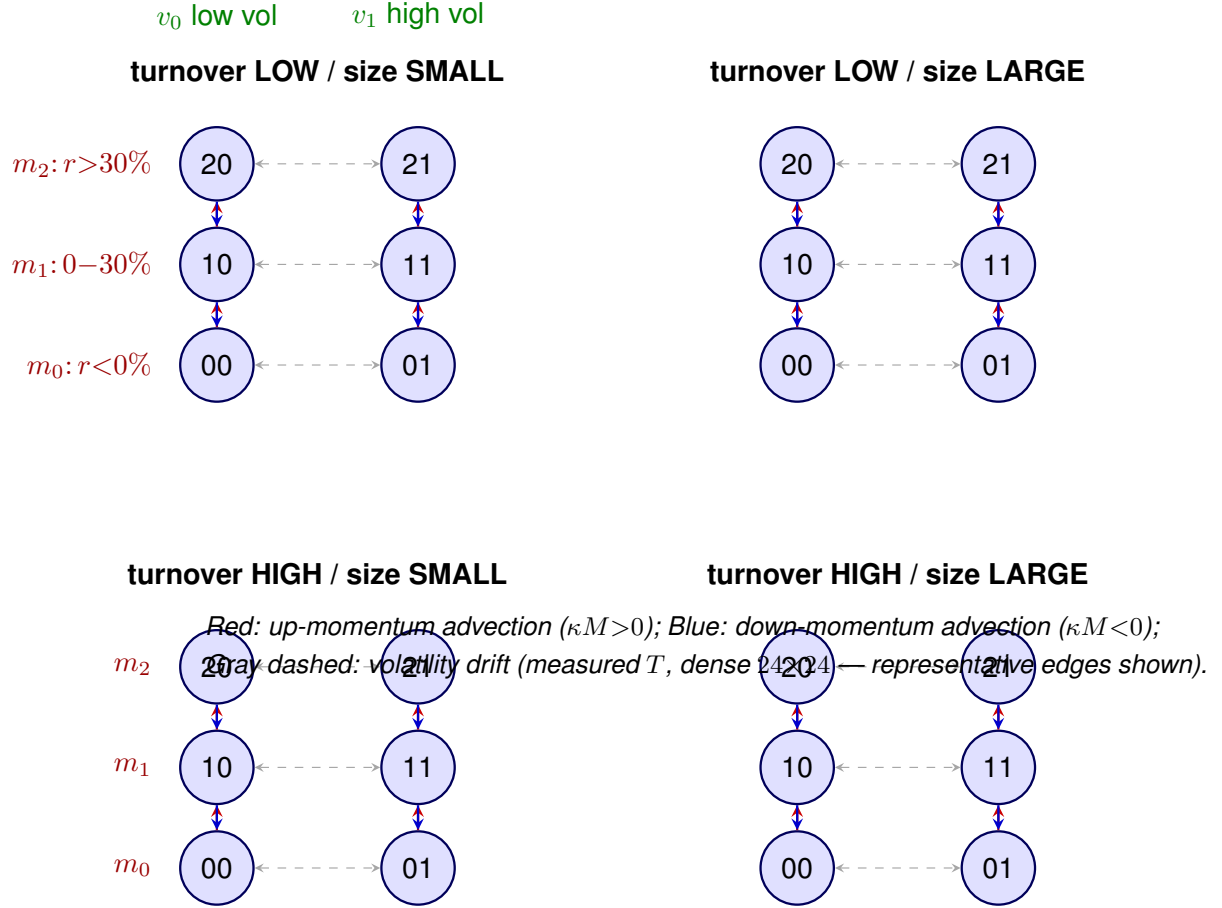


Figure 2: Reaction-admissibility graph: 24 bins (3 momentum $\times$ 2 volatility tiles, 2×2 by turnover $\times$ size). Red: up-momentum advection ($\kappa M > 0$); blue: down-momentum advection; gray dashed: measured volatility drift (representative edges of the dense T).

Table 3: Reaction kernels (rev-5 calibration on clean data).

Kernel	Form	Value	Label
Drift	$T_{\text{calm}}, T_{\text{stress}} (24 \times 24)$	measured	cap-weighted 2000–2007, clean pan
Market advection	$\kappa M_{\text{cs}}, w_{\text{eff}} = 0.15$	$\kappa = 1.0$	mechanical / assumed
Spike damper	$\min((M /B)^p, 1)$, gate $c > 0.5$	$p = 0.8, \tau = 6\text{mo}$	assumed (episode-tuned)
Herding h_1	bilinear toward winner lanes	0	calibrated (no gain on clean data)
Sticky drift λ_1	$T_{\text{eff}} = (1 - \lambda)T + \lambda I$	0	calibrated (no gain on clean data)

share between 0.999 and 0.002 month to month [10]. The repair (`fix_share_units.py`): per-ticker segmentation at $> 20 \times$ split-adjusted jumps, each segment checked against two absolute anchors — market cap in [\$200M, \$4T] and average-daily turnover in [0.02%, 100%] (dollar volume is **measured** [9]). 28 segments rescaled by $10^{\pm 3}, 10^{\pm 6}$; 17 dropped, never invented. Verification: total binnable cap 2007-12 **\$10.2T** (S&P 500 $\approx$ \$13T), 2021-12 **\$41.3T** ($\approx$ \$40T). Cap-weighting is multiplicative in the share count — on clean data the conditional hindcasts improve to RMSE **0.0899** (2008–09) and **0.0714** (2020–21), with the 2020–21 peak amplitude exact.

4.3 Scenario engine and fragility monitor

The forecastable object is $c(t)$, not returns. A 200-scenario ensemble from 2022-12 (Dirichlet-drawn T , **estimated** Kish N_{eff} 109–1825/bin; AR(1) market driver, **estimated**; VIX-regime Markov

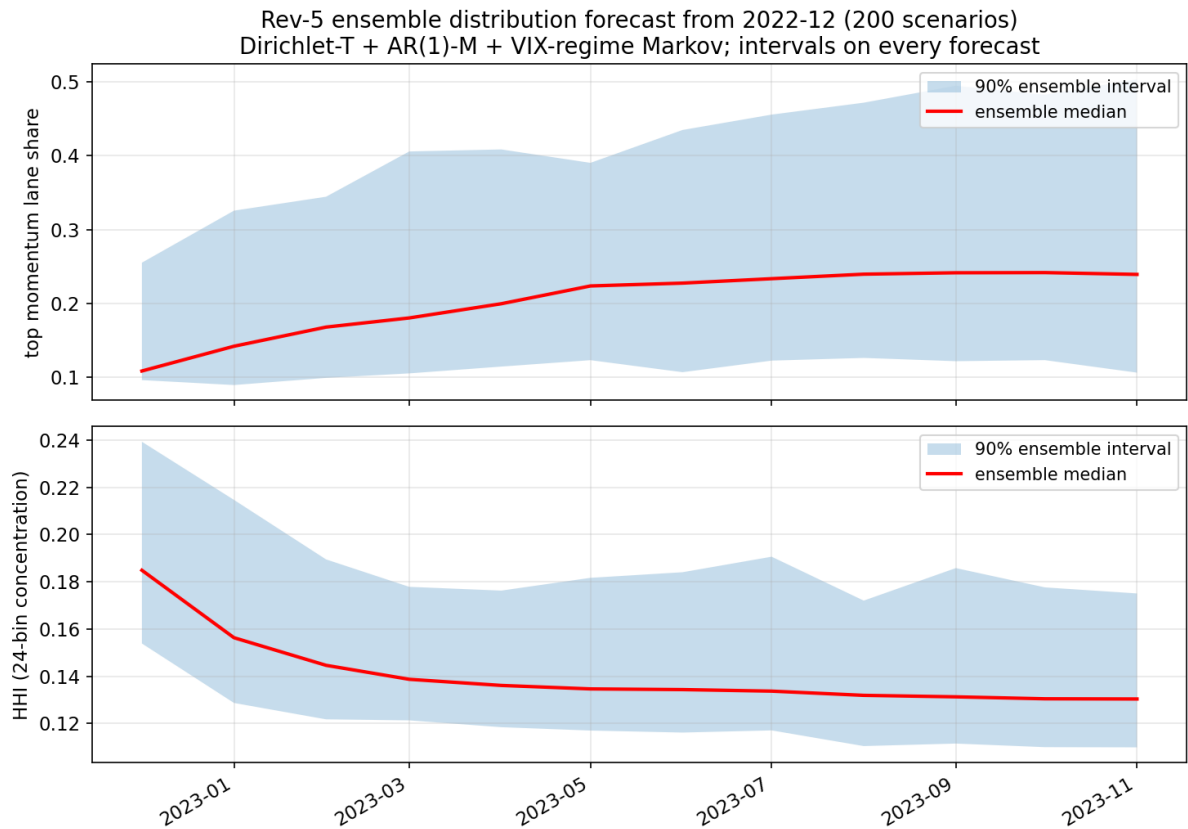


Figure 3: Ensemble distribution forecast from 2022-12 (200 scenarios): median and 90% intervals for top-lane share and HHI.

chain, **measured**) gives 12-month-ahead top-lane share **median 0.239, 90% interval [0.106, 0.503]**; HHI [0.110, 0.175] (Figure 3). The interval width is the finding: one-year concentration is dominated by driver and transition uncertainty.

The fragility monitor reports top-lane share $C(t)$ and HHI as percentiles of their own history (Figure 4). 2022-12: $C = 0.101$ (**25th percentile** — not crowded), HHI 52nd percentile. Most crowded month-ends: 2004-02 (0.570), 2010-01 (0.571), 2001-02 (0.423).

Standing verdict: the regime engine does not forecast returns and must not drive capital deployment. Its legitimate uses are regime census, conditional scenario analysis, and concentration/fragility monitoring.

5 Discarded candidates

Seven return-forecast mechanisms were tested with pre-registered designs and walk-forward validation. All failed; their full test documents remain in the repository as honest negatives. One-line reasons:

- **Rev-4 walk-forward tilt** — predicted mass-flow vs. realized bin return: correlation 0.009; tilt Sharpe 0.48 vs. SPY 0.64. Bin-boundary migration is reclassification, not price pressure.
- **Concentration → returns** — threshold-linear $t = -0.41$, walk-forward corr -0.045 ; quintile spreads statistically zero ($|t| < 0.6$), non-monotone. Honest negative.

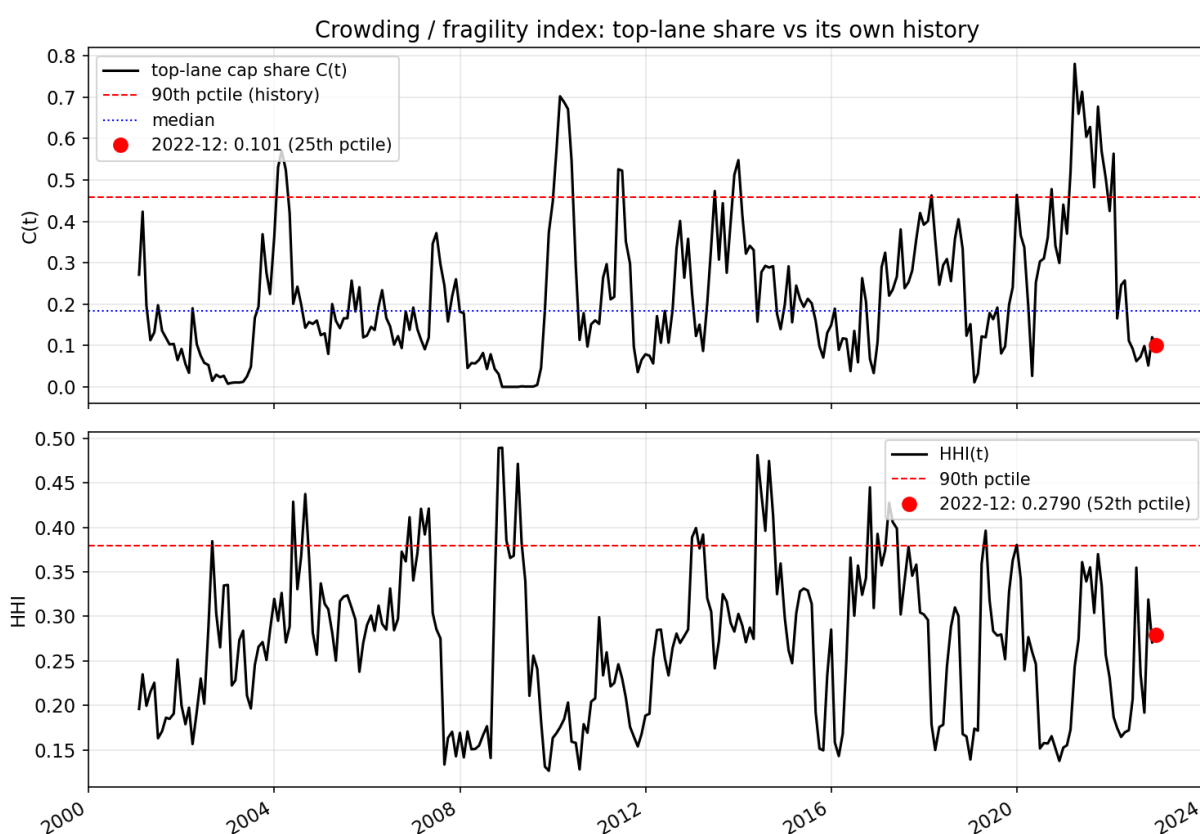


Figure 4: Fragility index: top-lane share and HHI vs. own history; 2022-12 readings marked. A state descriptor, not a signal.

- **Dispersion** → **returns** (Goyal–Santa-Clara style) — no replication in the large-cap panel; all out-of-sample R^2 negative.
- **Comomentum** (Lou–Polk) — winner-lane slope -0.46 (NW $t = -0.27$) with the wrong sign on the winner component; walk-forward corr -0.095 , OOS $R^2 = -0.082$.
- **Short interest** (Rapach–Ringgenberg–Zhou) — the published *negative* relationship reversed sign in the modern subsample (12-month $\beta = +0.42$, NW $t = +1.31$); not accepted post hoc.
- **Bubble characteristics** (Greenwood–Shleifer–You, adapted to individual stocks) — crash logit pseudo- $R^2 = 0.0134$; acceleration $p = 0.138$ with the wrong sign; issuance (one of the paper’s strongest predictors) unavailable. Weak evidence, not a refutation.
- **Hedge-fund crowding** (Brown–Howard–Lundblad) — public-13F proxy non-replication: all three pass criteria failed; the one marginal signal (hf_own 3-month, $t = +2.13$) collapsed to $t = +1.22$ under size control; CUSIP→ticker mapping covered only 25.6% of CUSIPs.

6 Next steps

1. **Paper-trade the Buffett screen** (frozen design, live tracking vs. SPY). Real capital only if the paper record holds; then a small pilot.
2. **Keep the regime engine as a diagnostic:** census, scenario intervals, fragility monitor. No return-forecast development without return-side economics (earnings, flows) — reclassification arithmetic is exhausted.

3. The discarded candidates' test documents stay in the repo as the permanent record of what was tried and why it was dropped.

References

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